Failure-Aware and Resource-Adaptive 2D Aerial Mosaicking from UAV Video

Diep Quang Sang, Nguyen Duc Nam, Dao Dang Nguyen Khoi, Truong Quang Dang Khoa
AI2002, Group 09
FPT University, Ho Chi Minh City, Vietnam

Abstract—Sequential 2D mosaicking of UAV video fails quietly: one bad pairwise registration is accumulated into the map and every later frame inherits the error. Classical matchers are cheap but brittle on low-texture terrain, while learned matchers are robust but too slow to run on every frame. We study three mechanisms on top of an incremental-homography pipeline: a cheap step-level risk indicator set, a two-tier matcher router, and three recovery policies, all measured against a full-resolution SIFT reference matcher on two 45-step windows of the NPU Drone-Map dataset. Three findings stand out. (i) Point-count indicators do not separate bad registrations at step level (AUC 0.40 for matches, 0.41 for inliers, below chance), whereas post-registration evidence does (AUC 0.77–0.86). (ii) Running the learned matcher on every step is both slower and worse than SIFT-only (60.4 s / 82.4 px vs 3.2 s / 75.3 px of final drift), while routing at τ_{ncc}=0.93 escalates 16% of steps and reaches 28.5 px, so the useful operating point is narrow rather than monotone. (iii) Under one shared trigger, frame-preserving submosaic splitting keeps all 46 frames, while frame dropping and keyframe re-matching keep only 14/46 because triggers cascade. Evidence is limited to two dev-selected windows, one dataset, and a matcher-consensus error proxy.

Index Terms—UAV video mosaicking, failure prediction, matcher routing, homography estimation, cumulative drift, recovery policy.

I. Introduction

Mosaicking a UAV flight into a single 2D image is a cheap alternative to full photogrammetric reconstruction when only a local situational picture is needed. The standard recipe is incremental: match frame i to frame $i-1$, estimate a homography H_i, accumulate it, and blend warped frames. The failure mode is also standard. Over low-texture or repetitive terrain the pairwise estimate degenerates, and because each step is conditioned on the previous one, a single bad H_i biases every subsequent frame (drift accumulation).

The literature offers two extremes. Handcrafted pipelines (SIFT/ORB with RANSAC [1], [2], [3]) are fast enough for CPU-only field use but break on hard frames; learned matchers (SuperPoint+LightGlue [4], [5], LoFTR variants [6], [7]) handle those frames but cost one to two orders of magnitude more compute, which matters when the whole flight must be processed on a laptop or in a Colab session. Existing incremental systems mostly accept or reject a frame using a fixed heuristic threshold [9], [10] rather than deciding how much compute a frame deserves, and the recovery behaviours that appear in individual systems (skip the frame, re-match an older keyframe, close a submosaic) have not, to our knowledge, been compared under one shared trigger on one pipeline.

This paper reports an empirical study of four questions on one incremental pipeline:

- RQ1 (failure prediction). Which cheap per-step indicators actually separate bad registrations from good ones?
- RQ2 (routing). What does a two-tier policy (classical by default, learned matcher only on flagged steps) buy in cost and in map quality, compared with always-classical and always-learned?
- RQ3 (recovery). Under one trigger, how do dropping the frame, re-matching a placed frame, and splitting the submosaic compare in frames kept and drift?

Our contributions are the measurements themselves, including two negative results that we consider the most transferable part of the study: inlier counts are anti-informative for step-level error, and calling the stronger matcher everywhere made the map worse than the cheap matcher on the window we measured.

II. Related Work

Incremental mosaicking. CPU-oriented sequential pipelines limit the number of features or apply a light inlier threshold to keep frame-to-frame registration affordable [9]; keyframe selection and weighted reprojection-error optimisation are used to damp drift [10]; seamline and warp selection address the blending side [12]; optical-flow based agricultural pipelines cluster frames into submosaics [11]. All of them expose heuristic accept/reject rules but not a cost-aware decision.

Matching confidence and failure prediction. In visual place recognition, inlier count and inlier ratio separate successful from failed matches, with false positives on repetitive texture [15], [16]. At homography level, large-scale benchmarks show that reprojection error and feature counts are weakly correlated with the final geometric error [17], [18]. Uncertainty estimation has been added to deep homography estimation for satellite–thermal geo-localisation [20], and model-vs-model consistency checks such as VSAC score inlier quality independently [19]. These works motivate cheap indicators but leave open which indicator predicts step error in a mosaicking chain.

Classical vs learned matching. SIFT+LightGlue was shown to recover low-texture and large-rotation UAV regions [13]; systematic comparisons on mobile-mapping and drone imagery find learnable methods more accurate but consistently slower [14]. Lightweight learned matchers (XFeat [8]), semi-dense matchers with token selection (Efficient LoFTR [7]) and UAV-specific attention designs [21] reduce but do not remove that cost. Our router is the classical-first counterpart of this trade-off, evaluated at mosaic level rather than per pair.

III. Method

A. Pipeline

Frames are read in temporal order. For step $(i-1, i)$ the pipeline extracts features, matches, estimates H_i with RANSAC, and accumulates $H_{i\to\text{root}}$. The cheap tier is SIFT + brute-force matching with Lowe's ratio test at max_side=640 and max_keypoints=2000. The strong tier is SuperPoint + LightGlue on the same 640-px images; an optional variant re-runs flagged steps at original resolution with the same matcher. Implementation: OpenCV 5.0.0, PyTorch 2.14.1 (CPU only), NumPy 2.4.6, RANSAC threshold 3 px, seed 0.

B. Reference-based step error

No field ground truth is available for these packages, so error is defined as disagreement with a reference estimate. For every step the reference matcher (full-resolution SIFT, 4000 features, same RANSAC) produces H_i^{ref}; steps whose reference solution has fewer than 100 inliers are marked invalid. The step error is the mean displacement of five marker points (four corners plus centre of the source frame, expressed in original-image coordinates) between the pipeline homography and H_i^{ref}:

$$e_i = \frac{1}{5}\sum_{k=1}^{5} \left\| H_i x_k - H_i^{\text{ref}} x_k \right\|_2 . \tag{1}$$

A step is bad when $e_i > 2$ px, which is the measured noise floor of the reference matcher itself on this data. Chain drift is reported separately as the accumulated difference between the pipeline chain and the reference chain, in pixels.

C. Indicators and trigger

Eight cheap indicators are logged per step: matches, inliers, inlier_ratio, rmse_px (internal RANSAC RMSE), hull_area_ratio (spatial spread of inliers), overlap_ratio (projected frame intersection), photo_ncc (normalised cross-correlation of the aligned pair) and photo_bad_frac (share of pixels deviating more than 25 grey levels after alignment). Every indicator has a suspicion direction fixed before evaluation (e.g. high photo_bad_frac is suspicious, low photo_ncc is suspicious), so no direction is chosen after seeing labels. A step is flagged when photo_ncc $< \tau$ or inlier_ratio < 0.30 or no homography was produced. τ is selected by rule, not by eye: the largest value such that both evaluation windows stay within a 25% escalation budget.

D. Recovery policies

Four policies run on the same pipeline and the same trigger: none (register anyway), drop (skip the frame), anchor (re-match the nearest already-placed frame instead of the previous frame), and split (close the current submosaic and start a new one at the frame). Recovery is measured by frames placed, number of submosaics, number of chained steps, and the maximum internal drift of each chain.

IV. Experimental Setup

Data. NPU Drone-Map [22] public keyframe packages (undistorted 1920×1080 RGB, ∼400 frames per package). Two packages are used here: phantom3-village-kfs (hard window = frames 80–125, where SIFT disagrees with the reference by 5–11 px) and phantom3-grass-kfs (frames 0–45, where Efficient LoFTR disagrees by 5–313 px). Windows were picked after a scan of 160 steps per package; the first 20 frames were excluded because all seven matcher configurations stitched all 20 of them without stopping, leaving no failure case to predict. Each window is 46 frames / 45 steps. Artifacts include SHA-256 hashes of all 92 input frames, the environment manifest, and per-step records; the full notebook ran on CPU in 507 s. The mosaic renderings are produced by reports/paper/make_mosaics.py, which replays the stored homographies without re-running any matcher and asserts that the drift of every rebuilt chain equals the tabulated value.

Metrics. For RQ1, Spearman ρ between indicator and step error and ROC-AUC for the bad/good step label, using the pre-declared direction; AUC is reported only when both classes are non-empty. For RQ2, cumulative matcher time (seconds, matcher stage only), escalation rate, mean step error and final drift. For RQ3, frames placed, submosaic count, chained steps, and maximum internal drift.

V. Results

A. RQ1: what predicts a bad step

Table I lists the village/SIFT window, where 21 of 45 steps are bad. The two point-count indicators are the worst of the set: for both, the declared direction (fewer matches or inliers is suspicious) is inverted by the data, giving AUC 0.402 (matches) and 0.409 (inliers), i.e. steps with more matches were slightly more likely to be bad. Post-registration evidence dominates: photometric disagreement after warping (0.855), internal RMSE (0.839), overlap (0.788) and NCC (0.770). The same ordering holds for SuperPoint+LightGlue on the same window (27/45 bad; photo_ncc 0.798, photo_bad_frac 0.757, matches 0.436) and on grass/SuperPoint+LightGlue (6/38 bad; photo_ncc 0.865, matches 0.615). Two windows have no defined AUC: grass/SIFT has zero bad steps (0/38 valid steps), and Efficient LoFTR on village is bad on 45/45 steps (the diagnostic case, not a learnable signal).

TABLE I
RQ1, village/SIFT+BF window: 45 steps, 21 labelled bad. Direction is the pre-declared suspicion sign. AUC is computed with that sign, while Spearman ρ uses raw indicator values, so an AUC below 0.5 marks an indicator whose declared direction is reversed in this window.

Indicator	Direction	Spearman ρ	AUC
photo_bad_frac	↑	+0.759	0.855
rmse_px	↑	+0.788	0.839
overlap_ratio	↓	−0.634	0.788
photo_ncc	↓	−0.636	0.770
inlier_ratio	↓	−0.389	0.726
hull_area_ratio	↓	+0.005	0.571
inliers	↓	+0.239	0.409
matches	↓	+0.251	0.402



Fig. 1. Mosaics of the village window (46 frames, forward flight), rebuilt from the stored homographies: (a) SIFT only, (b) routed at τ_{ncc}=0.93. The strip silhouette is the warped frame boundary; the difference in drift (75.3 px vs. 28.5 px) is measured, not read off the picture.

B. RQ2: cost and quality of routing

Table II and Fig. 3(b) show the trade-off on village. Always-learned is the worst of the three policies on both axes: it costs 18.7× the matcher time of SIFT-only and ends 9% further from the reference chain. Routing at τ_{ncc}=0.93 escalates 7 of 45 steps (15.6%, inside the 25% budget) and reduces final drift from 75.3 px to 28.5 px, i.e. 62% lower, at 4.1× SIFT-only matcher time. Escalating those same seven steps at original resolution further reduces step-level error (1.49 px vs 4.51 px on the flagged steps) but ends with larger chain drift (35.8 px), which we discuss in Sec. VI.

On grass the picture differs: SIFT is the accurate matcher there (0 bad steps against the reference; final drift 14.7 px in 2.4 s), while SuperPoint +LightGlue costs 49.0 s and drifts 38.4 px. Consequently the budget rule selects τ=0.96 and escalates nothing; the router correctly declines to spend compute on this window. Fig. 1 and Fig. 2 show the maps those policies produce.

C. RQ3: recovery under one trigger

Table III reports both windows. On village (7 initial triggers), split places all 46 frames across 8 submosaics, chaining 38 of 45 steps with a maximum internal drift of 21.7 px; drop and anchor keep only 14/46 frames (13 chained steps, the same 21.7 px ceiling), because dropping one frame enlarges the overlap gap of the next pair and triggers the guard again (7 initial triggers cascade to 32). anchor performs identically to drop here: with every remaining candidate either already-rejected or unplaced, no safe anchor frame existed. Fig. 5 shows what the split policy actually produces.

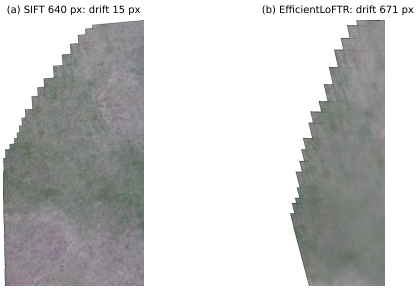


Fig. 2. Same 46 frames of grass, two matchers, no recovery: (a) SIFT at 640 px, (b) Efficient LoFTR. The staircase shear on the right is the accumulated 671.3 px disagreement with the reference chain, on terrain where the learned matcher reports dense, low-residual (but wrong) matches.

TABLE II
RQ2, village window (45 steps). "Strong" = steps sent to SuperPoint+LightGlue; time is matcher-stage time only.

Policy	Strong	Time (s)	e_i mean (px)	Drift (px)
SIFT only	0/45	3.2	2.51	75.3
SP+LG only	45/45	60.4	2.67	82.4
Routed τ=0.93	7/45	13.3	2.42	28.5
Routed + full-res	7/45	73.6	1.95	35.8

On grass with Efficient LoFTR, 44 of 45 steps are flagged. drop and anchor place a single frame (the map does not survive); split keeps all 46 frames but degenerates into 45 one-frame fragments with a single chained step. Low drift there is an artefact of that fragmentation, not evidence of a recovered map.

VI. Discussion

Cheap does not mean informative. Match and inlier counts, the indicators named in most acceptance heuristics, carried no usable signal at step level in our window, and the data inverted their declared direction (fewer matches is suspicious): steps with many matches can still be placed consistently with a wrong geometry. Indicators computed after registration (photometric agreement, internal residual, projected overlap) carry the signal. The practical cost is that they cannot be evaluated before matching, so our trigger is a post-hoc guard applied to the cheap matcher's output, not a pre-matching predictor.

Spending more compute is not monotone in quality. Always-learned ended worse than SIFT-only on village, and per-step accuracy improved by escalating at full resolution did not translate into lower chain drift. The plausible mechanism is that the accumulated map is a product of the whole chain, not of the average step: a

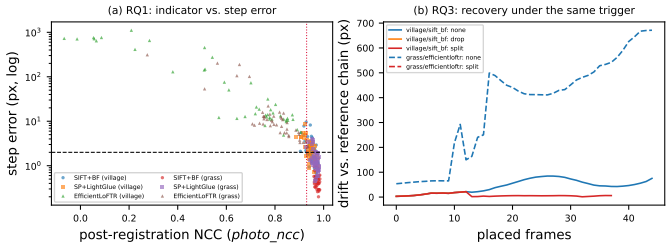


Fig. 3. (a) Step error against the post-registration NCC indicator on both windows and all three matchers; dashed line = 2 px bad/good boundary, dotted line = the selected trigger. (b) Drift against the reference chain for RQ3 policies under the same trigger. Split resets the coordinate frame at each fragment, so its curve is piecewise, not a single chain.

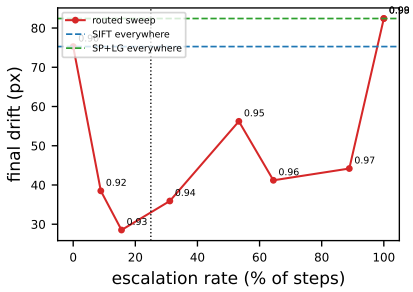


Fig. 4. RQ2 threshold sweep on village: escalation rate against final drift. Horizontal lines are the two fixed policies; the dotted line is the 25% escalation budget. Drift is non-monotone in τ, so the operating point is chosen by budget, not by "higher threshold = safer".



Fig. 5. Split policy on village: the same 46 frames, cut at the 7 flagged steps into 8 fragments, each in its own coordinate frame. All frames survive, but the downstream consumer receives 8 disconnected surfaces, four of which (fragments 3, 5, 6 and 8) are single frames.

TABLE III
RQ3: recovery policies at the RQ2-selected trigger (village τ=0.93, grass τ=0.96). Drift is the maximum internal drift within a chain; split resets the frame per fragment, and n/a means no frame was placed, so there is nothing to measure.

Window / matcher	Policy	Frames kept	Fragments	Drift (px)
village / SIFT	none	46/46	1	84.6
village / SIFT	drop	14/46	1	21.7
village / SIFT	anchor	14/46	1	21.7
village / SIFT	split	46/46	8	21.7
grass / ELoFTR	none	46/46	1	671.3
grass / ELoFTR	drop	1/46	1	n/a
grass / ELoFTR	split	46/46	45	2.7

locally better-conditioned H_i changes which frame geometries are compensated later. We treat this as a hypothesis; testing it requires varying the escalation set independently of the trigger.

Threshold selection is the hard part. Fig. 4 shows the sweep is U-shaped: 15.6% escalation is the best point measured, while escalating half the steps (53%) is worse than escalating none. A budget rule ("largest τ such that escalation stays below 25%") produced 0.93/0.96 on the two windows, but a rule that assumes monotonicity over a range where the metric is non-monotone is only as good as the windows it was fitted on.

Recovery trade-off is frames vs. connectivity. Split preserves the frame inventory but fragments the map (8 pieces for 46 frames, 45 pieces in the degenerate case); drop/anchor preserve continuity of the surviving chain but cascade under a conservative trigger, losing 70% of frames here. Which is preferable depends on whether the downstream use needs coverage or a single geometric surface; a question our numbers cannot settle, and one reason the mosaic images for each policy should be inspected before drawing conclusions.

VII. Limitations

Error proxy. All errors are disagreement with a full-resolution SIFT reference chain, which itself accumulates error; nothing here is validated against field geometry (no GCP/orthophoto comparison in this draft), so absolute drift values should not be read as georeferencing accuracy.

Sample. Two of the 14 public NPU Drone-Map packages, two windows chosen because they are hard. Base rates of bad steps (21/45, 6/38, 45/45) are properties of that selection and must not be read as deployment expectations. There is no held-out sequence yet, and the trigger and budget have not been re-tested on the remaining packages.

Selection and multiplicity. Indicators, direction, the 2 px boundary and τ were all chosen while looking at these windows; the direction was fixed in advance to remove the most obvious leak, but the results remain exploratory rather than confirmatory. The 45 steps are also temporally dependent, not 45 independent samples.

Cost accounting. The time column measures the matcher stage only. RANSAC, warping, the photometric checks and the risk decision are excluded, so this draft supports no end-to-end latency claim; the true routing overhead is higher than 13.3 s and the true always-learned overhead also includes its own post-processing.

Integration. RQ3 runs a fixed matcher per window and does not use the routes produced in RQ2; an end-to-end "cheap → escalate → recover" run, its mosaic outputs, and per-fragment connectivity measures are still missing. All timings come from one CPU-only machine.

VIII. Conclusion

On an incremental-homography UAV pipeline, the cheap indicators that predict useful step-level risk are not the point counts that acceptance heuristics usually use but post-registration evidence; a classical-first router with a budget-selected trigger cut final drift by 62% at 4.1× the cheap matcher's time while calling the learned matcher everywhere was slower and worse; and under one shared trigger, frame-preserving submosaic splitting kept all frames where drop/anchor cascaded to 14/46. Next steps, in order: re-run the same notebook unchanged on held-out packages, integrate routing with recovery into one end-to-end run with full-pipeline timing, validate against field geometry, and inspect the per-policy mosaics.

References

[1] D. G. Lowe, "Distinctive image features from scale-invariant keypoints," Int. J. Comput. Vis., vol. 60, no. 2, pp. 91–110, 2004.

[2] E. Rublee, V. Rabaud, K. Konolige, and G. Bradski, "ORB: An efficient alternative to SIFT or SURF," in Proc. IEEE Int. Conf. Comput. Vis. (ICCV), 2011, pp. 2564–2571.

[3] M. A. Fischler and R. C. Bolles, "Random sample consensus: A paradigm for model fitting with applications to image analysis and automated cartography," Commun. ACM, vol. 24, no. 6, pp. 381–395, 1981.

[4] D. DeTone, T. Malisiewicz, and A. Rabinovich, "SuperPoint: Self-supervised interest point detection and description," in Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. Workshops (CVPRW), 2018, pp. 224–236.

[5] P. Lindenberger, P.-E. Sarlin, and M. Pollefeys, "LightGlue: Local feature matching at light speed," in Proc. IEEE/CVF Int. Conf. Comput. Vis. (ICCV), 2023.

[6] J. Sun, Z. Shen, Y. Wang, H. Bao, and X. Zhou, "LoFTR: Detector-free local feature matching with transformers," in Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR), 2021, pp. 8922–8931.

[7] Q. Wang et al., "Efficient LoFTR: Semi-dense local feature matching with sparse-like speed," in Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR), 2024.

[8] P. Potlapalli, P. Lindenberger, and M. Pollefeys, "XFeat: Accelerated features for lightweight image matching," in Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR), 2024.

[9] D. Hwang et al., "Real-time 2D orthomosaic mapping using CPU-based incremental homography matrix estimation with UAV video," Appl. Sci., vol. 16, no. 4, p. 2133, 2026.

[10] X. Li et al., "A real-time incremental video mosaic framework for UAV remote sensing," Remote Sens., vol. 15, no. 8, p. 2127, 2023.

[11] M. Kharismawati and D. Kazic, "DroneZaic: A robust end-to-end pipeline for mosaicking freely flown aerial video of agricultural fields," Plant Phenome J., vol. 8, no. 1, p. e70033, 2025.

[12] J. Chen et al., "UAV video stitching based on optimal seamline and half-projective warp," Remote Sens., vol. 14, no. 5, p. 1068, 2022.

[13] J. Kim and S. Kim, "Robust UAV image mosaicking using SIFT and LightGlue," Int. Arch. Photogramm. Remote Sens. Spatial Inf. Sci., vol. XLVIII-2-W11-2025, pp. 169–173, 2025.

[14] S. Gaisbauer, P. Gyawali, Q. Zhang, O. Wysocki, and B. Jutzi, "To glue or not to glue? Classical vs learned image matching for mobile mapping cameras to textured semantic 3D building models," ISPRS Ann. Photogramm. Remote Sens. Spatial Inf. Sci., vol. X-1/W2-2025, 2025.

[15] M. Zaffar et al., "On the estimation of image-matching uncertainty in visual place recognition," in Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR), 2024.

[16] C. Sferrazza et al., "To match or not to match: Evaluating keypoint matching confidence for visual place recognition," in Proc. CVPRW, 2024.

[17] D. Barath et al., "A large-scale homography benchmark," in Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR), 2023.

[18] H. Yuan, D. Peng, and M. Dong, "Comparative analysis of color space transformations on feature-based image registration," J. Imaging, vol. 10, no. 5, p. 105, 2024.

[19] M. Ivashechkin, D. Barath, and J. Matas, "VSAC: Efficient and accurate estimator for H and F," in Proc. IEEE/CVF Int. Conf. Comput. Vis. (ICCV), 2021.

[20] J. Xiao and G. Loianno, "UASTHN: Uncertainty-aware deep homography estimation for UAV satellite–thermal geo-localization," arXiv:2502.01035, 2025.

[21] H. Zhang, H. He, Y. Zhou, and F. Lei, "Efficient image matching for UAV visual navigation via DALGlue," Sci. Rep., vol. 15, p. 37684, 2025.

[22] S. Bu, Y. Zhao, G. Wan, and Z. Liu, "Map2DFusion: Real-time incremental UAV image mosaicing based on monocular SLAM," in Proc. IEEE/RSJ Int. Conf. Intell. Robots Syst. (IROS), 2016.